

Psoriasis: A Comprehensive Clinical Summary

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- Psoriasis: A Comprehensive Clinical Summary
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2025 Edition

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Psoriasis: A Comprehensive Clinical Summary

Overview: Psoriasis — A Chronic Inflammatory Systemic Disease

Psoriasis is a genetically determined, chronic inflammatory systemic autoimmune disease affecting approximately 1–2% of the global population. It occurs with equal prevalence in males and females and can manifest at any age, with a median age of onset at 28 years. Two distinct peaks of onset are recognized: an early peak between ages 16–22 and a late peak between ages 57–60. Importantly, psoriasis is not merely a cutaneous condition — it is a systemic disease with significant implications for overall health, including associations with seronegative arthritis, cardiovascular disease, metabolic syndrome, type 2 diabetes, obesity, depression, and substantially reduced quality of life.

KEY TEACHING POINTS

Psoriasis is a systemic inflammatory disease, not just a skin condition — always screen for comorbidities.

The two peaks of onset (early: 16–22; late: 57–60) reflect potentially different pathogenic triggers and clinical phenotypes.

Up to 50% of children develop psoriasis if both parents are affected, underscoring the strong genetic component.

The DLQI (Dermatology Life Quality Index) is the standard tool for assessing quality of life impact in psoriasis patients.

Pathophysiology: The Triad of Psoriasis Development

The pathogenesis of psoriasis involves three interconnected mechanisms: genetic predisposition, environmental triggers, and immune dysregulation. Understanding this triad is essential for comprehending both the clinical presentation and the rationale behind modern targeted therapies.

1. Genetic Predisposition

Psoriasis has a well-established hereditary component. The PSORS1 locus on chromosome 6 is the most significant genetic susceptibility region, and specific HLA associations — particularly HLA-Cw6 — are strongly linked to early-onset disease. The familial risk is substantial: if one parent is affected, approximately 16% of children will develop psoriasis; if both parents are affected, this risk rises to 50%. In monozygotic twins, concordance reaches 70%, compared to only 20% in dizygotic twins, confirming that while genetics play a major role, environmental factors are also critical.

2. Environmental Triggers

Multiple environmental factors can initiate or exacerbate psoriasis in genetically susceptible individuals:

- Streptococcal pharyngitis is a classic trigger for guttate psoriasis; HIV infection can also precipitate or worsen psoriasis.
- Physical injury to the skin can induce new psoriatic lesions at the site of trauma — the Koebner phenomenon.
- Psychological stress is a well-recognized trigger for flares.

- Beta-blockers, antimalarials (e.g., hydroxychloroquine), lithium, and abrupt withdrawal of systemic corticosteroids can all trigger or worsen psoriasis.
- Smoking and obesity are independent risk factors for development and severity.
- Pregnancy can either improve or worsen psoriasis.
- Hypocalcemia can precipitate pustular psoriasis.

3. Immune Dysregulation

The central immunological driver of psoriasis involves T-helper lymphocytes, particularly the Th1 and Th17 pathways. Activated T cells produce a cascade of pro-inflammatory cytokines — most notably TNF-alpha, IFN-gamma, IL-17, and IL-23 — which act on keratinocytes, driving their rapid proliferation and perpetuating the inflammatory cycle. This cytokine network is the primary target of modern biologic therapies.

MCQ TRAPS & PITFALLS

Do NOT confuse the Th1/Th17 immune pathways of psoriasis with the Th2-dominant pathways seen in atopic dermatitis.

HLA-Cw6 is associated with early-onset psoriasis, not late-onset disease.

Abrupt withdrawal of systemic corticosteroids (not initiation) triggers pustular psoriasis flares.

Pathophysiology: Keratinocyte Hyperproliferation and Histological Hallmarks

Under normal physiological conditions, keratinocytes originate in the basal layer and slowly migrate to the skin surface over approximately 23 days. In psoriasis, this cell turnover is dramatically accelerated, with cells reaching the surface in only 3–5 days. This rapid transit results in accumulation of cells at the surface (hyperkeratosis) and failure of proper maturation. Normally, keratinocytes lose their nuclei as they ascend through the epidermis; however, in psoriasis, cells move so rapidly that they retain their nuclei throughout the epidermis — a histological feature known as **parakeratosis**.

The rapid turnover and failure of proper maturation results in defective keratinocytes that are poorly adherent and easily scraped off, revealing underlying dilated blood vessels — this is the **Auspitz sign**. The inflammatory infiltrate in the dermis leads to edema and erythema, while collections of inflammatory cells at the skin surface can form sterile pustules, most commonly seen in palmoplantar pustular psoriasis.

KEY TEACHING POINTS

Normal epidermal turnover: ~23 days; Psoriatic epidermal turnover: 3–5 days.

Parakeratosis (retained nuclei in the stratum corneum) is the hallmark histological feature of psoriasis.

Auspitz sign: pinpoint bleeding when scales are scraped off, exposing dilated dermal capillaries.

Munro microabscesses (collections of neutrophils in the stratum corneum) are another key histological feature.

Clinical Presentation: Recognizing Psoriatic Lesions

Psoriasis is characterized by well-demarcated erythematous plaques with overlying silvery scales. The erythema results from dilated dermal blood vessels, while the silvery scaling reflects rapid keratinocyte

proliferation. The most common locations are the knees and elbows — sites prone to minor trauma (Koebner phenomenon). Lesions typically improve with sun exposure, and psoriasis is usually only mildly pruritic.

The main clinical features reflect the underlying pathological processes:

Plaques: Well-defined raised areas of psoriasis, the cardinal lesion.

Scaling: Silvery-white scales due to rapid keratinocyte turnover; easily scraped off revealing the Auspitz sign.

Erythema: Results from dilated dermal capillaries in the elongated dermal papillae.

Koebner phenomenon: New psoriatic lesions developing at sites of skin trauma, including surgical scars, sunburn, or scratches.

Nail changes: Pitting (small depressions in the nail plate), onycholysis (separation of the nail from the nail bed), subungual hyperkeratosis, and oil-drop discoloration (salmon patches).

Clinical Features of Psoriatic Plaques and Their Pathological Basis

Clinical Feature	Pathological Basis	Diagnostic Significance
Well-demarcated plaques	Localized keratinocyte hyperproliferation	Distinguishes psoriasis from eczema (poorly demarcated)
Silvery scales	Parakeratosis and hyperkeratosis	Characteristic; not seen in most other dermatoses
Auspitz sign	Dilated capillaries in elongated papillae exposed by scale removal	Highly suggestive but not pathognomonic
Koebner phenomenon	Trauma-induced T-cell activation and cytokine release	Seen in active disease; also in lichen planus and vitiligo
Nail pitting	Psoriatic involvement of the nail matrix	Present in up to 50% of psoriasis patients

Classification of Psoriasis Subtypes

Psoriasis manifests in several distinct clinical subtypes, each with characteristic morphology, distribution, and clinical significance. Accurate classification guides treatment decisions and prognostic assessment.

1. Chronic Plaque Psoriasis (Psoriasis Vulgaris)

This is the most common form, accounting for approximately 85–90% of all psoriasis cases. It presents as well-demarcated, erythematous plaques with silvery-white scales, typically distributed symmetrically on extensor surfaces (knees, elbows), scalp, sacral region, and gluteal cleft. Lesions may range from a few small plaques to widespread involvement covering most of the body surface area. The severity is classified as mild (<3% BSA), moderate (3–10% BSA), or severe (>10% BSA) using body surface area assessment, or by the Psoriasis Area and Severity Index (PASI) score.

2. Guttate Psoriasis

Guttate psoriasis presents as numerous small (1–10 mm), drop-shaped, erythematous papules and plaques with fine scale, distributed predominantly on the trunk and proximal extremities. It is most common in children and young adults and is classically triggered by group A beta-hemolytic streptococcal pharyngitis, typically appearing 2–3 weeks after the infection. The condition may be self-limiting, resolving within 3–4 months, but approximately 40–60% of patients may eventually develop chronic plaque psoriasis.

3. Pustular Psoriasis

Pustular psoriasis is a rare but potentially life-threatening variant characterized by sterile pustules on an erythematous base. It is subdivided into:

Generalized pustular psoriasis (von Zumbusch type): Widespread eruption of sterile pustules with systemic toxicity — fever, malaise, leukocytosis. This is a dermatological emergency requiring hospitalization. Triggers include abrupt withdrawal of systemic corticosteroids, hypocalcemia, infections, and pregnancy.

Palmoplantar pustulosis: Chronic sterile pustules localized to the palms and soles, often associated with smoking and thyroid disease.

Acrodermatitis continua of Hallopeau: Rare variant affecting the fingertips and nails, potentially leading to nail destruction and bone changes.

4. Erythrodermic Psoriasis

Erythrodermic psoriasis is the rarest but most dangerous subtype, characterized by generalized erythema and scaling affecting >90% of body surface area. It results from uncontrolled inflammation leading to widespread vasodilation and increased skin blood flow, causing significant fluid loss, thermoregulatory failure, and high-output cardiac failure. Patients are at risk for sepsis, hypoalbuminemia, and electrolyte imbalances. This is a dermatological emergency requiring immediate hospitalization.

5. Inverse (Flexural) Psoriasis

Inverse psoriasis affects intertriginous areas — axillae, groin, inframammary folds, and the gluteal cleft. Lesions are typically well-demarcated, bright red, and shiny with minimal scaling due to the moist environment. It is often misdiagnosed as candidal intertrigo or seborrheic dermatitis. Friction and sweating exacerbate the condition.

6. Scalp Psoriasis

Scalp involvement occurs in approximately 50–80% of psoriasis patients and is often the first site of presentation. It presents as well-demarcated, erythematous plaques with silvery scale, which may extend beyond the hairline. It can be the most difficult form to treat due to the presence of hair limiting topical therapy application.

Psoriasis Subtypes — Clinical Features and Key Associations

Subtype	Morphology	Distribution	Key Associations
Chronic plaque (vulgaris)	Well-demarcated erythematous plaques with silvery scale	Extensor surfaces, scalp, sacrum	Most common (85–90%); Koebner phenomenon
Guttate	Small drop-shaped papules with fine scale	Trunk, proximal extremities	Streptococcal pharyngitis; children/young adults
Pustular (generalized)	Sterile pustules on erythematous base with systemic toxicity	Widespread	Abrupt steroid withdrawal; dermatological emergency
Erythrodermic	Generalized erythema and scaling >90% BSA	Entire body surface	Cardiac failure, fluid loss, sepsis risk

Subtype	Morphology	Distribution	Key Associations
Inverse (flexural)	Bright red, shiny, minimally scaling plaques	Axillae, groin, inframammary folds	Misdiagnosed as candidal intertrigo
Scalp	Erythematous plaques with silvery scale extending beyond hairline	Scalp	Present in 50–80% of patients; often first site

Psoriatic Arthritis

Psoriatic arthritis (PsA) is a chronic inflammatory arthritis associated with psoriasis, classified under the seronegative spondyloarthropathies. It affects approximately 20–30% of psoriasis patients, and in about 15% of cases, the arthritis precedes the skin disease. The condition is defined by the absence of rheumatoid factor (seronegative) and is associated with HLA-B27 in certain subtypes.

Classification: The CASPAR Criteria

The Classification Criteria for Psoriatic Arthritis (CASPAR) are used for diagnosis. A patient must have inflammatory articular disease (joint, spine, or enthesal) with ≥ 3 points from the following features:

- Current psoriasis (2 points) or personal/family history of psoriasis (1 point)
- Psoriatic nail dystrophy (pitting, onycholysis) — 1 point
- Negative rheumatoid factor — 1 point
- Current or history of dactylitis — 1 point
- Radiographic evidence of juxta-articular new bone formation — 1 point

Clinical Patterns of Psoriatic Arthritis

Five classical patterns of PsA are recognized:

- 1. Asymmetric oligoarthritis (most common — 40–50%):** Involvement of ≤ 4 joints, often in a "sausage digit" pattern (dactylitis). Typically affects DIP joints of hands and feet.
- 2. Symmetric polyarthritis (25%):** Resembles rheumatoid arthritis but is seronegative. Affects ≥ 5 joints symmetrically, commonly MCP, PIP, and wrist joints.
- 3. Distal interphalangeal (DIP) joint predominant (10%):** Predominantly affects DIP joints with prominent nail changes (pitting, onycholysis). Highly specific for PsA.
- 4. Spondylitis (5%):** Sacroiliitis and spinal involvement, often asymmetric. Associated with HLA-B27. May be indistinguishable from ankylosing spondylitis.
- 5. Arthritis mutilans (rare — <5%):** Severe, destructive form with bone resorption leading to "pencil-in-cup" deformity on radiographs and telescoping of digits (opera-glass hand).

KEY TEACHING POINTS

Psoriatic arthritis is seronegative — rheumatoid factor is negative.

Dactylitis ("sausage digit") and enthesitis are hallmark features distinguishing PsA from RA.

Nail changes (pitting, onycholysis) are strongly associated with DIP joint involvement.

Up to 15% of PsA patients develop arthritis before skin disease — always ask about joint symptoms in psoriasis patients.

HLA-B27 is associated with the spondylitis subtype, not all PsA.

Comorbidities: Psoriasis as a Systemic Disease

The systemic inflammatory nature of psoriasis drives a wide range of comorbidities that significantly impact morbidity and mortality. Recognition and screening for these comorbidities is an essential component of comprehensive psoriasis management.

Cardiovascular Disease

Psoriasis — particularly moderate-to-severe disease — is an independent risk factor for cardiovascular disease. The chronic systemic inflammation accelerates atherosclerosis, increasing the risk of myocardial infarction, stroke, and peripheral vascular disease. Patients with severe psoriasis have a 50% increased risk of cardiovascular mortality and may have a life expectancy reduced by approximately 3–5 years. The risk is highest in young patients with severe disease.

Metabolic Syndrome

Psoriasis is strongly associated with metabolic syndrome, including obesity, type 2 diabetes mellitus, dyslipidemia, and hypertension. The prevalence of metabolic syndrome in psoriasis patients is approximately 40%, compared to 20–25% in the general population. Obesity is both a risk factor for developing psoriasis and a consequence of the disease (due to reduced physical activity, depression, and social isolation).

Psoriatic Arthritis

As discussed above, 20–30% of psoriasis patients develop psoriatic arthritis, which can cause progressive joint destruction and functional disability if untreated.

Depression and Psychosocial Impact

The visible nature of psoriasis, chronicity of disease, and social stigma contribute to high rates of depression, anxiety, and social isolation. Approximately 20–30% of psoriasis patients experience clinically significant depression. The DLQI (Dermatology Life Quality Index) scores in severe psoriasis are comparable to those seen in cancer and cardiovascular disease.

Other Comorbidities

- **Inflammatory bowel disease:** Increased risk of Crohn's disease (3-fold increased risk).
- **Uveitis:** Associated with psoriatic arthritis, particularly the spondylitis subtype.
- **Non-alcoholic fatty liver disease (NAFLD):** Increased prevalence, particularly in patients on long-term methotrexate.
- **Chronic kidney disease:** Moderately increased risk.

- **Lymphoma:** Slightly increased risk, though confounded by immunosuppressive therapy.

IMPORTANT

Psoriasis is an independent cardiovascular risk factor — do NOT attribute CV risk solely to traditional risk factors.

Screen ALL psoriasis patients for psoriatic arthritis at every visit — early detection prevents irreversible joint damage.

Depression screening is essential — the psychosocial burden of psoriasis is comparable to cancer and heart disease.

Psoriasis Comorbidities — Screening Recommendations

Comorbidity	Prevalence in Psoriasis	Screening Recommendation
Cardiovascular disease	50% increased mortality risk in severe disease	Annual BP, lipid panel, glucose; assess CV risk factors
Metabolic syndrome	~40%	BMI, waist circumference, fasting glucose, lipid panel
Psoriatic arthritis	20–30%	Ask about joint pain, stiffness, dactylitis at every visit
Depression	20–30%	PHQ-9 or DLQI screening annually
Inflammatory bowel disease	3-fold increased risk (Crohn's)	Ask about GI symptoms; refer if suspected
NAFLD	Increased	LFTs, especially if on methotrexate
Uveitis	Associated with PsA spondylitis subtype	Ocular symptoms inquiry; ophthalmology referral if symptomatic

Diagnosis of Psoriasis

Clinical Diagnosis

Psoriasis is primarily a clinical diagnosis based on the characteristic morphology (well-demarcated erythematous plaques with silvery scale), distribution (extensor surfaces, scalp, sacrum), and associated features (nail pitting, Auspitz sign, Koebner phenomenon). A thorough history should include:

- Duration and progression of skin lesions
- Family history of psoriasis (present in ~30% of patients)
- Recent streptococcal infection (guttate psoriasis)
- Medication history (beta-blockers, lithium, antimalarials)
- Joint symptoms (screen for psoriatic arthritis)
- Psychosocial impact (DLQI assessment)
- Comorbidity screening (cardiovascular risk factors, metabolic syndrome)

Dermoscopy

Dermoscopy of psoriatic plaques reveals uniformly distributed dotted vessels (glomerular vessels) on a white scaly background with a reddish halo. This vascular pattern is highly characteristic and helps differentiate psoriasis from other scaly dermatoses.

Histopathology

Skin biopsy is rarely needed for typical presentations but may be performed when the diagnosis is uncertain. Key histological features include:

- **Parakeratosis:** Retained nuclei in the stratum corneum (hallmark feature)
- **Munro microabscesses:** Collections of neutrophils in the stratum corneum
- **Regular acanthosis:** Elongation of rete ridges with uniform elongation ("test-tube" appearance)
- **Thinning of the suprapapillary plates:** Thinning of epidermis overlying dilated dermal papillae
- **Dilated capillaries in elongated dermal papillae:** Correlates with Auspitz sign
- **Spongiform pustules of Kogoj:** Neutrophil collections in the stratum spinosum

Laboratory Investigations

No specific laboratory test confirms psoriasis. Investigations are primarily used to:

- Screen for comorbidities (FBC, LFTs, renal function, fasting glucose, lipid panel)
- Assess for streptococcal infection (ASO titer, throat culture — if guttate psoriasis suspected)
- Baseline before systemic therapy (FBC, LFTs, renal function, hepatitis B/C serology, TB screening)
- Rule out differential diagnoses (KOH scraping to exclude tinea, syphilis serology if indicated)

OSCE PEARLS

In OSCE skin stations, always examine the nails, scalp, and extensor surfaces — these are the most commonly affected sites.

Ask about joint symptoms in every psoriasis patient — examiners expect screening for psoriatic arthritis.

Mention the Koebner phenomenon when describing psoriasis — it demonstrates understanding of disease pathophysiology.

Always assess psychosocial impact — mention DLQI in your management plan.

Management of Psoriasis

The management of psoriasis is guided by disease severity, subtype, patient preference, and comorbidities. A stepwise approach is used, escalating from topical therapy to phototherapy to systemic agents and biologics.

Assessment of Severity

Disease severity determines the treatment pathway:

- **Mild psoriasis:** <3% BSA involvement, PASI <10, DLQI <10 — managed with topical therapy
- **Moderate psoriasis:** 3–10% BSA, PASI 10–20, DLQI 10–20 — consider phototherapy or systemic therapy
- **Severe psoriasis:** >10% BSA, PASI >20, DLQI >20, or involvement of high-impact sites (hands, feet, face, genitalia) — systemic therapy or biologics

Topical Therapy (First-Line for Mild Disease)

Emollients: First-line for all patients. Reduce scaling, improve skin barrier function, and enhance penetration of active topical agents. Should be applied liberally and frequently.

Topical corticosteroids: Mainstay of topical treatment for mild-to-moderate psoriasis. Potency should be matched to body site:

- Face and flexures: Mild potency (e.g., hydrocortisone 1%)
- Trunk and limbs: Moderate-to-potent (e.g., betamethasone valerate 0.1%)
- Scalp, palms, soles: Potent-to-very potent (e.g., betamethasone dipropionate 0.05%)

Side effects include skin atrophy, striae, telangiectasia, and rebound flare on withdrawal. Use the fingertip unit (FTU) to guide dosing — 1 FTU = 0.5g, covers approximately 2% BSA.

Vitamin D analogues (calcipotriol/calcitriol) Normalize keratinocyte proliferation and differentiation. Used as monotherapy or in combination with topical corticosteroids (e.g., Enstilar foam, Daivobet). Safe for long-term use; avoid in hypercalcemia. Maximum dose: 100g/week for adults.

Coal tar: Reduces scaling and inflammation. Available in shampoos, creams, and ointments. Effective for scalp psoriasis. Limitations include odor, staining, and messiness.

Topical calcineurin inhibitors (tacrolimus, pimecrolimus): Useful for facial and flexural psoriasis where corticosteroid side effects are a concern. Not first-line but valuable for sensitive areas.

Dithranol (anthralin): Effective for stable plaque psoriasis. Applied as short-contact therapy (10–30 minutes then washed off). Stains skin and clothing; can irritate surrounding skin.

Phototherapy (Second-Line for Moderate Disease)

Narrowband UVB (NB-UVB, 311 nm): First-line phototherapy for moderate-to-severe psoriasis. Effective, safe, and widely available. Administered 2–3 times per week; typical course requires 20–30 sessions. Long-term risk of photoaging and skin cancer.

PUVA (Psoralen + UVA): Oral or topical psoralen followed by UVA exposure. More effective than NB-UVB but with greater long-term risks (skin cancer, cataracts). Reserved for severe or recalcitrant disease. Requires eye protection for 24 hours after oral psoralen.

Systemic Therapy (Third-Line for Moderate-to-Severe Disease)

Methotrexate: First-line systemic agent for moderate-to-severe psoriasis and psoriatic arthritis. Dose: 7.5–25 mg once weekly (oral or subcutaneous). Mechanism: inhibits dihydrofolate reductase, reducing DNA synthesis and immune cell proliferation. Requires monitoring: FBC, LFTs, renal function every 2–4 weeks initially, then every 3 months. Folic acid 5mg weekly (not on methotrexate day) reduces side effects. Hepatotoxicity risk with cumulative dose — consider liver biopsy or FibroScan if cumulative dose >1.5g. Contraindicated in pregnancy (teratogenic).

Ciclosporin: Calcineurin inhibitor with rapid onset of action. Dose: 2.5–5 mg/kg/day in two divided doses. Used for severe psoriasis, erythrodermic psoriasis, and as a "bridge" therapy while waiting for slower-acting

agents to take effect. Maximum duration: 1–2 years due to nephrotoxicity risk. Requires monitoring: BP, renal function, electrolytes, lipids. Contraindications: uncontrolled hypertension, renal impairment, malignancy.

Acitretin: Oral retinoid. Dose: 10–50 mg daily. Particularly effective for pustular and erythrodermic psoriasis. Often combined with PUVA (Re-PUVA) for enhanced efficacy. Side effects: cheilitis, dry mucous teratogenicity (contraindicated in pregnancy; avoid conception for 3 years after discontinuation in women). Monitor: LFTs, lipids.

Dimethyl fumarate (Skilarence): Licensed for moderate-to-severe plaque psoriasis. Mechanism involves immunomodulation via the Nrf2 pathway. Side effects: GI symptoms (diarrhea, abdominal pain), flushing, lymphopenia. Requires FBC monitoring.

Biologic Therapy (For Moderate-to-Severe Disease Refractory to Conventional Systemics)

Biologic agents target specific components of the immune system and have revolutionized the treatment of moderate-to-severe psoriasis. They are indicated when conventional systemic therapy is ineffective, contraindicated, or not tolerated.

TNF-alpha inhibitors:

- Etanercept (Enbrel): 25–50 mg SC twice weekly or 50 mg weekly
- Infliximab (Remicade): 5 mg/kg IV at weeks 0, 2, 6, then every 8 weeks
- Adalimumab (Humira): 80 mg SC loading, then 40 mg every 2 weeks
- Certolizumab pegol (Cimzia): 400 mg SC at weeks 0, 2, 4, then 200 mg every 2 weeks

IL-12/23 inhibitor:

- Ustekinumab (Stelara): 45–90 mg SC at weeks 0, 4, then every 12 weeks

IL-23 inhibitors (p19 subunit):

- Guselkumab (Tremfya): 100 mg SC at weeks 0, 4, then every 8 weeks
- Risankizumab (Skyrizi): 150 mg SC at weeks 0, 4, then every 12 weeks
- Tildrakizumab (Ilumya): 100 mg SC at weeks 0, 4, then every 12 weeks

IL-17 inhibitors:

- Secukinumab (Cosentyx): 300 mg SC at weeks 0, 1, 2, 3, 4, then monthly
- Ixekizumab (Taltz): 160 mg SC loading, then 80 mg every 2 weeks for 12 weeks, then every 4 weeks
- Brodalumab (Siliq): 210 mg SC at weeks 0, 1, 2, then every 2 weeks
- Bimekizumab (Bimzelx): 320 mg SC at weeks 0, 4, 8, 12, 16, then every 8 weeks

IMPORTANT

All biologic patients must be screened for latent TB (chest X-ray, IGRA) and hepatitis B/C before starting therapy.

Live vaccines are contraindicated during biologic therapy.

TNF-alpha inhibitors carry a risk of demyelinating disease — avoid in patients with multiple sclerosis.

IL-17 inhibitors may exacerbate inflammatory bowel disease — avoid in Crohn's disease.

Biologics are NOT first-line — they are indicated after failure of conventional systemic therapy.

Biologic Therapies for Psoriasis — Mechanism and Key Considerations

Biologic Agent	Target	Route	Key Considerations
Adalimumab	TNF-alpha	SC	Most widely used; TB reactivation risk; self-administered
Infliximab	TNF-alpha	IV	Rapid onset; infusion reactions; used in acute/severe disease
Secukinumab	IL-17A	SC	High efficacy (PASI 90 ~70%); candidiasis risk
Ixekizumab	IL-17A	SC	Highest PASI 90 rates (~73%); injection site reactions
Guselkumab	IL-23 (p19)	SC	Dosing every 8 weeks; sustained response; favorable safety
Risankizumab	IL-23 (p19)	SC	Dosing every 12 weeks; high PASI 90 (~75%)
Ustekinumab	IL-12/23 (p40)	SC	Long track record; weight-based dosing; every 12 weeks

Management of Special Situations

Pregnancy: Preferred agents include topical corticosteroids (mild-moderate potency), emollients, and NB-UVB phototherapy. Methotrexate and acitretin are absolutely contraindicated (teratogenic). Certolizumab pegol is the only biologic considered safe in pregnancy (no placental transfer due to Fc-free structure).

HIV-associated psoriasis: May be severe and refractory. Antiretroviral therapy often improves psoriasis. Use systemic agents cautiously due to immunosuppression. Phototherapy may be useful.

Erythrodermic and pustular psoriasis: Dermatological emergencies requiring hospitalization. Fluid and electrolyte management, temperature regulation, and treatment of superinfection. Ciclosporin or infliximab for rapid control; acitretin for pustular variants.

Prognosis and Monitoring

Psoriasis is a chronic, relapsing condition with no cure. The disease course is unpredictable, with periods of remission and flare. Long-term monitoring should include:

- Regular assessment of disease severity (PASI, BSA, DLQI)
- Annual screening for comorbidities (cardiovascular risk factors, metabolic syndrome, depression)
- Monitoring for psoriatic arthritis (clinical assessment at every visit)
- Medication-specific monitoring (LFTs for methotrexate, renal function for ciclosporin, FBC for dimethyl fumarate)
- Assessment of treatment response and adjustment of therapy as needed

The treat-to-target approach aims for PASI 90 or absolute PASI ≤ 2 , reflecting the high efficacy achievable with modern biologic therapies.

MCQ TRAPS & PITFALLS

Methotrexate is given ONCE WEEKLY, not daily — daily dosing causes fatal bone marrow suppression.

Acitretin requires 3 years of contraception after discontinuation in women (not 1 year like isotretinoin).

Ciclosporin is nephrotoxic — monitor renal function, not just LFTs.

IL-17 inhibitors (secukinumab, ixekizumab) can exacerbate IBD — avoid in Crohn's disease.

Certolizumab pegol is the safest biologic in pregnancy due to lack of placental transfer.

Practice Questions

Q: A 24-year-old woman presents with a 2-week history of numerous small, drop-shaped, erythematous papules on her trunk and proximal arms. She had a sore throat 3 weeks ago that resolved without treatment. On examination, the lesions have fine scale and are distributed predominantly on the trunk. What is the most likely diagnosis?

- A. Pityriasis rosea
- B. Guttate psoriasis
- C. Secondary syphilis
- D. Lichen planus
- E. Tinea corporis

Answer: B

Explanation: Guttate psoriasis classically presents in young adults 2–3 weeks after streptococcal pharyngitis with numerous small, drop-shaped erythematous papules with fine scale on the trunk and proximal extremities. Pityriasis rosea has a herald patch and Christmas-tree distribution. Secondary syphilis involves the palms and soles. Lichen planus presents with pruritic, violaceous, polygonal papules. Tinea corporis presents with annular, scaly plaques with central clearing.

Q: A 45-year-old man with a 10-year history of plaque psoriasis presents with progressive pain and swelling of his right index finger and left knee for 3 months. On examination, the right index finger is diffusely swollen ("sausage digit") and there is tenderness at the Achilles tendon insertion. Nail examination reveals pitting and onycholysis. Rheumatoid factor is negative. What is the most likely diagnosis?

- A. Rheumatoid arthritis
- B. Gout
- C. Psoriatic arthritis
- D. Reactive arthritis
- E. Osteoarthritis

Answer: C

Explanation: The combination of pre-existing psoriasis, dactylitis (sausage digit), enthesitis (Achilles tendon tenderness), nail changes (pitting, onycholysis), and negative rheumatoid factor is classic for psoriatic arthritis. Rheumatoid arthritis is seropositive and does not cause dactylitis. Gout typically presents with acute monoarticular inflammation (1st MTP). Reactive arthritis follows genitourinary or gastrointestinal infection. Osteoarthritis does not cause dactylitis or enthesitis.

Q: A 38-year-old woman with severe plaque psoriasis (PASI 25, DLQI 18) has failed treatment with methotrexate and ciclosporin. She is being considered for biologic therapy. Which of the following is the most important investigation before initiating treatment?

- A. Echocardiogram
- B. Chest X-ray and interferon-gamma release assay (IGRA)
- C. Colonoscopy
- D. Bone marrow biopsy
- E. Lumbar puncture

Answer: B

Explanation: Before initiating biologic therapy, screening for latent tuberculosis (chest X-ray and IGRA) is mandatory, as biologics — particularly TNF-alpha inhibitors — can reactivate latent TB. Hepatitis B/C screening is also required. Echocardiogram is not routinely required. Colonoscopy, bone marrow biopsy, and lumbar puncture are not indicated for biologic screening.

Q: A 52-year-old man presents with generalized erythema and scaling affecting 95% of his body surface area. He is febrile (38.8°C), tachycardic (HR 110), and hypotensive (BP 90/60). He has a history of chronic plaque psoriasis and recently stopped taking oral prednisolone after a 3-week course. What is the most likely diagnosis and immediate management?

- A. Cellulitis — start IV antibiotics
- B. Erythrodermic psoriasis — hospitalize for fluid resuscitation and temperature regulation
- C. Drug eruption — discontinue all medications
- D. Staphylococcal scalded skin syndrome — IV flucloxacillin
- E. Toxic epidermal necrolysis — transfer to burns unit

Answer: B

Explanation: Erythrodermic psoriasis is a dermatological emergency characterized by generalized erythema and scaling affecting >90% BSA, with systemic toxicity (fever, tachycardia, hypotension). The most common trigger is abrupt withdrawal of systemic corticosteroids. Management requires hospitalization, fluid and electrolyte management, temperature regulation, and systemic therapy (ciclosporin or infliximab). The clinical history of recent steroid withdrawal strongly supports this diagnosis.

Q: A 30-year-old woman with moderate plaque psoriasis is planning to become pregnant. Which of the following medications is considered the safest option for use during pregnancy?

- A. Methotrexate
- B. Acitretin
- C. Adalimumab

- D. Certolizumab pegol
- E. Ciclosporin

Answer: D

Explanation: Certolizumab pegol is the only biologic considered safe in pregnancy because it lacks an Fc region and therefore does not cross the placenta via active transport. Methotrexate and acitretin are absolutely contraindicated (teratogenic). Adalimumab crosses the placenta (especially in the third trimester) and should be avoided or timed carefully. Ciclosporin may be used in severe cases but is not the safest option.

Q: A 60-year-old man with psoriasis presents with well-demarcated, bright red, shiny plaques in his axillae and groin with minimal scaling. What is the most likely subtype of psoriasis?

- A. Chronic plaque psoriasis
- B. Guttate psoriasis
- C. Inverse (flexural) psoriasis
- D. Pustular psoriasis
- E. Erythrodermic psoriasis

Answer: C

Explanation: Inverse (flexural) psoriasis affects intertriginous areas (axillae, groin, inframammary folds) and presents with bright red, shiny, well-demarcated plaques with minimal scaling due to the moist environment. It is often misdiagnosed as candidal intertrigo. Chronic plaque psoriasis has prominent silvery scale on extensor surfaces. Guttate psoriasis presents with drop-shaped papules on the trunk. Pustular psoriasis has sterile pustules. Erythrodermic psoriasis involves >90% BSA.

Q: A patient with psoriasis is being treated with methotrexate 15 mg weekly. Which of the following is the most appropriate monitoring schedule?

- A. FBC and LFTs every 6 months
- B. FBC and LFTs every 2–4 weeks initially, then every 3 months
- C. Renal function only, every 3 months
- D. No monitoring required at this dose
- E. Chest X-ray annually

Answer: B

Explanation: Methotrexate requires regular monitoring of FBC (for bone marrow suppression) and LFTs (for hepatotoxicity). The recommended schedule is every 2–4 weeks during dose titration and initial treatment, then every 3 months once stable. Renal function should also be monitored. A baseline chest X-ray is obtained, but annual chest X-ray is not required for monitoring. Folic acid 5 mg weekly (on a different day from methotrexate) reduces side effects.